

Mentored Research Grant (MRG)

Cover Page

Project Title

Slow Actuation of Nitinol Stents Using Focused Ultrasound

Principal Investigator(s) (full-time faculty)

PI Name: Dr. Christopher R. Dillon

PI Department or Program: Mechanical Engineering

Co-PI names and departments (if applicable): Dr. Jeffrey R. Hill, Mechanical Engineering

Abstract

Rapidly deployed stents used to treat peripheral artery disease can cause acute injury to blood vessels leading to restenosis. Such stents are often made from Nitinol, a shape memory alloy with the ability to change shape when heated above its transition temperature. Focused ultrasound is an emerging therapeutic technology for non-invasively heating tissues deep within the body. This project will merge these technologies, investigating the potential of focused ultrasound to non-invasively and slowly actuate Nitinol stents to expand peripheral arteries without causing damage. Undergraduate researchers will carry out the work under the guidance of several faculty and graduate students. Through this process, they will enhance their skills in conducting experiments, reviewing literature, design, writing, and presenting their findings. Additionally, they will have the chance to mentor other undergraduate students as part of their training.

Number of Students to be Mentored

Number of Undergraduate Students: 2

Number of Graduate Students*: 1

* MRGs are primarily for mentoring undergraduates in research. When graduate students are included in the budget, it should be in support of creating the mentoring environment.

Budget

Provide the budget in the table below. The total amount can be up to \$25,000. MRGs do not cover tuition, equipment, or faculty wages.

Item	Year 1	Year 2	Total
Undergraduate wages	\$10000	\$8500	\$18500
Graduate wages	\$1500	\$1500	\$3000
Supplies	\$1000	\$500	\$1500
Travel		\$2000	
Other (explain)			
Total	\$12500	\$12500	\$25000

Explain "Other" if used:

Relationship to External Funding

While Dr. Dillon has expertise in focused ultrasound and Dr. Hill in smart materials, combining these research areas represents a new direction for each of them. This MRG will be used as seed funding to generate preliminary data for a NIH R15 grant proposal to develop a focused ultrasound-actuated, slowly deployable stent for the treatment of peripheral artery disease.

Research Motivation and Overview

Peripheral artery disease (PAD) is characterized by narrowing of arteries in the arms or legs and associated reductions in blood flow to the extremities due to buildup of fatty plaque on arterial walls. PAD affects more than 6.5 million adults over 40 years of age in the United States [1]. Chronic pain, aching, or cramps when walking, muscle atrophy, and sores or ulcers that do not heal are just a few of the ways that PAD negatively impact patients' quality of life [2].

Endovascular stenting is a common treatment for the narrowed blood vessels of peripheral arterial disease (PAD). It is widely used due to its minimally invasive nature, as well as the associated reduction in short-term morbidity and mortality rates [3]. However, the rapid expansion of stents during placement to expand disease-narrowed arteries can injure the blood vessel walls. The body's response to this acute injury is called neointimal hyperplasia and leads to in-stent restenosis (ISR), the gradual re-narrowing of a stented artery [4]. This research explores a new method to slowly expand Nitinol stents through the application of focused ultrasound, with the goal of reducing ISR caused by stent insertion.

The expected outcomes of this work are three-fold: 1) Provide a research experience for two undergraduate students that includes mentoring opportunities from multiple faculty and graduate students as well as the chance to train and mentor other undergraduates, 2) Publish peer-reviewed articles with multiple student co-authors for two young faculty expanding into a new research area, 3) generate preliminary data for a NIH R15 proposal that will lead to the development of a focused ultrasound-actuated, slowly deployable stent for the treatment of PAD.

Background

Nitinol is a shape memory alloy (SMA) made of nickel and titanium that is known for its unique mechanical properties. Specifically, its shape memory effect and superelasticity make it ideal for a wide range of applications in fields like biomedical engineering, aerospace, and robotics [5]. Nitinol can "remember" its original shape and return to it when exposed to a specific temperature, which is known as its transition temperature. This is due to the reversible martensitic transformation that occurs when Nitinol is heated or cooled through a specific temperature range. Nitinol can be "set" to a specific shape or configuration through thermal training. Thermal training involves heating the Nitinol high above its transition temperature, deforming it into the desired shape, and then cooling it while maintaining the deformation. Once cooled, the Nitinol can undergo significant deformation and still recover its original shape, once it is heated above its transition temperature.

There are two common types of stents used in medical procedures: plastically deforming stents made of materials like stainless steel, and self-expanding stents made of Nitinol [3]. In both types, a small balloon (angioplasty) is used to pre-dilate the area of arterial stenosis, expanding the closed artery. In plastically deforming stents, the stent is located and then plastically deformed in a similar manner via balloon angioplasty and remains rigid to hold open the artery.

Self-expanding stents are slightly more complicated. The stents are designed and set to their final shape in the austenite (hot) phase, and then cooled to the martensite phase. Once cooled, the stent can be deformed to its insertion configuration, which is normally then constrained by a removable outer sheath. After insertion, the sheath is removed and the intrinsic hyper-elastic properties of the material allows it to rapidly deploy against the surrounding arterial walls [6].

Both types of stents suffer from adverse effects resulting from the rapid expansion of the stent, namely in-stent restenosis (ISR). ISR is induced by neointimal hyperplasia, the thickening of the blood vessel walls in response to vessel injury caused by the rapid stent expansion. ISR occurs in approximately 20% of angioplasty patients, and a majority of these patients require additional procedures within 6 months [7] [8].

Ultrasound is a diagnostic tool for visualizing soft tissues deep within the body using high frequency sound waves. If the power is increased and the sound waves are focused, as a magnifying glass focuses light, ultrasound can also heat up tissue for therapeutic effects. The heating is completely non-invasive and localized to a region as small as a grain of rice. Therapeutic focused ultrasound (FUS) has been used medically to increase blood flow and encourage tissue healing, locally release drugs to tumors through thermally sensitive liposomes, enhance the effects of radiation and chemotherapy, instigate immunotherapeutic responses to disease, ablate and destroy cancer, dissolve blood clots, and destroy diseased nerves [9]. There are hundreds of clinical applications for FUS at various stages of research, development, and commercialization [10].

This research investigates the previously unexplored use of focused ultrasound to slowly and precisely heat Nitinol stents for a controlled expansion and deployment that will not damage vessel walls and reduce the likelihood of ISR.

Preliminary ex vivo demonstration

A current undergraduate research assistant working with Dr. Dillon and Dr. Hill has investigated the feasibility of using focused ultrasound to heat Nitinol wire above its transition temperature within a reasonable time frame, while also avoiding excessive heating of the surrounding tissue. To examine this, straight Nitinol wire was bent to a 90-degree angle and placed between two layers of excised pork muscle. Different powers of focused ultrasound were subsequently applied to determine the duration required to heat the wire and initiate actuation. In addition, thermocouples recorded the temperature of the pork muscle at various distances from the focus's center. Figure 1 illustrates the experimental setup. Very early results have demonstrated that Nitinol wire can be heated in situ using focused ultrasound without causing permanent damage to nearby tissue.

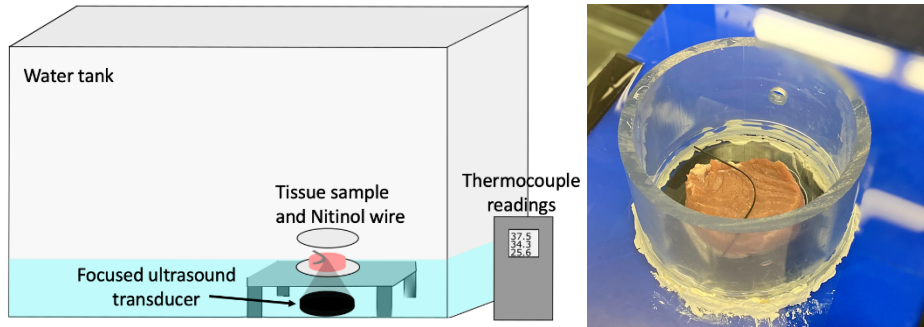


Figure 1. Left diagram shows the setup for Nitinol actuation in the water tank. Right image shows the actual experiment with the top pork muscle removed to show the bent Nitinol.

Student Research Project 1

While preliminary data show promise for the use of focused ultrasound in Nitinol actuation, more careful experimentation is warranted. The current configuration utilizes Nitinol wire with a transition temperature of 30 °C (too low to be used in vivo) with the surrounding pork muscle at room temperature (instead of the nominal body temperature of 37 °C). Our initial setup also neglects the potential impact of blood flow on focused ultrasound-mediated Nitinol actuation. As blood flows past the stent, it will draw thermal energy deposited by the ultrasound away from the stent, potentially preventing the stent from reaching its transition temperature for actuation.

The first undergraduate research assistant will investigate the parameters under which in vivo Nitinol actuation might be achieved. This work will utilize wire that transitions at 40 °C, and the surrounding “tissue” will be preheated to 37 °C through immersion in a temperature-controlled water bath. To mimic in vivo conditions, a silicone tube approximately the diameter of the iliac artery (~10 mm) will be embedded in a tissue mimicking gelatin-based phantom [11]. A body-temperature blood surrogate will be driven through the tubing from a peristaltic pump to replicate physiological flow rates. The Nitinol wire will be wrapped into a coiled spring shape, inserted into the tubing, and heated from an external focused ultrasound transducer. When the Nitinol exceeds its transition temperature, the wire will actuate. As it “unwinds”, the coil will expand, and the wire will lodge itself in the tubing. A variety of focused ultrasound powers and flow rates will be tested, since the angioplasty balloons used for arterial pre-dilation may restrict blood flow. Thermocouples will be placed in the tissue-mimicking phantom in several locations to assess the safety of nearby tissues during the heating process.

In addition to the experimental aspects of this project, the undergraduate research assistant will perform a literature search for ultrasound absorbent coatings or other techniques that might be utilized to increase the heating rate of the wire. The student will prepare an abstract and present findings at the Utah Conference for Undergraduate Research (**UCUR**) in 2024 and at the Biomedical Engineering Society (**BMES**) 2025 Annual Conference. The student will prepare and submit a manuscript for publication to *Physics in Medicine and Biology* on the non-invasive actuation of Nitinol using focused ultrasound.

Student Research Project 2

The second undergraduate research assistant will concentrate on designing and optimizing a Nitinol stent which can be expanded in situ. As previously discussed, existing stents open rapidly, frequently leading to ISR in the artery. Our primary objective is to develop the framework which will be necessary to design, model, and optimize a Nitinol stent assuming focused ultrasound actuation. For example, we want to be able to minimize the speed at which the stent opens, and so our design criteria must consider the rate of opening. The student will follow the design process used by BYU Capstone [12] to determine design criteria, explore concepts, and develop a prototype stent. Inspiration will be derived from present commercial stent designs and past research at BYU. Previously, the Babel Lab produced a multi-stage, gradually deploying stent which employs bioresorbable links to restrict stent expansion [13]. The links deteriorate at varying rates, allowing the stent to open slowly. Although this technique varies from our proposal, we will consult with Dr. Bowden to integrate relevant design concepts.

We do not expect a fully functional, optimized stent to be developed during this initial work, but we do expect to have a preliminary prototype which can be used to show the feasibility of the method for future funded work. The student will prepare an abstract and present findings at the Utah Conference for Undergraduate Research (**UCUR**) in 2024 and present the preliminary prototype at the ASME Smart Materials, Adaptive Structures, and Intelligent Systems (**SMASIS**) 2025 Annual Conference. A manuscript describing the device design and study findings will be submitted for publication to *Biomaterials Research*.

Mentoring environment

Both student research projects proposed for this mentored research grant are ideal for undergraduates- attention to detail and consistent effort will stimulate progress without requiring graduate-level knowledge or coursework. The undergraduate research assistants will **meet weekly with Dr. Dillon and Dr. Hill** to provide progress updates, set goals, and address problems. For more hands-on guidance, one of Dr. Dillon's **graduate students** (whose full-time project is in a different area, but who would like more mentoring experience) **will train the undergraduates** in equipment use and lab best practices and supervise their day-to-day work. A portion of the budget is allocated for this graduate student to ensure that they can take the time to contribute to the mentoring of these students.

The two student research projects are distinct, but interrelated, such that the research assistants will work on parallel paths while needing to team and communicate consistently with each other and their faculty mentors. In the second year of the project, each **research assistant will be paired with a freshman or sophomore student** (potentially on research mentorships from the BE Together program), **whom they will train** to assist in their project as it moves toward completion and publication.

Both research assistants will attend and participate in Dr. Dillon's **research lab group meetings**. The weekly meetings begin with a 5-10 minute life lesson prepared by Dr. Dillon on topics ranging from seeking internships and improving writing to leading with kindness and seeking the Spirit's influence. In addition to updates and advice for individual projects, each meeting has two student presenters. The first presenter reviews an article from the scientific literature, sharing the most important lab-relevant methods and findings in 5-10 minutes. The second presenter prepares a 15-minute PowerPoint deep dive into their current research project. The opportunity for students to learn to **read the scientific literature**, **succinctly summarize** what they learned, **prepare presentations** on their work, and **present their work** in front of others will prepare them for their engineering careers and is an important part of the mentoring environment.

Expected research outcomes

- The results of these studies will be submitted as abstracts for **four conference presentations** at the 2024 UCUR, 2025 ASME SMASIS Meeting, and 2025 Biomedical Engineering Society Annual Meeting. The students will lead the abstract and presentation development, attend the conferences, and present their work.
- The student projects will result in **two peer-reviewed journal articles** to be submitted to *Biomaterials Research* and *Physics in Medicine and Biology*. These publications will include multiple student co-authors and enable Dr. Dillon and Dr. Hill to establish footing in a new research area that merges their areas of core expertise.
- For new faculty members, it is challenging to generate preliminary data required to successfully acquire external funding. These two projects will provide preliminary data for a **NIH Research Enhancement Award** (R15, \$300k over 3 years) **proposal** submitted to the National Heart, Lung, and Blood Institute (NHLBI) by Dr. Dillon and Dr. Hill in June of 2025. This proposal will lead to the development of a focused ultrasound-actuated, slowly deployable stent for the treatment of PAD.

Potential impact of work

Working with the BYU Technology Transfer Office, the design efforts of these studies and subsequent research projects could lead to a patent for a novel focused ultrasound-actuated stent. With PAD being an extremely prevalent disease, the elimination or even mitigation of intimal hyperplasia and restenosis of arteries through a slowly deployable, non-invasively actuated stent has enormous potential for impacting the health and quality of life for millions of people.

The merging of these two distinct and emerging fields of research (focused ultrasound and shape memory alloys) could lead to future investigation of previously unconsidered and unexplored biomedical applications. For example, the ability to actuate Nitinol wire utilizing focused ultrasound may prove useful for other biomedical devices where external power sources are required, such as in the control of steerable needles for tissue biopsies [14].

References

- [1] S. S. Virani *et al.*, “Heart Disease and Stroke Statistics—2021 Update,” *Circulation*, vol. 143, no. 8, pp. e254–e743, 2021, doi: 10.1161/CIR.0000000000000950.
- [2] J. W. Olin and B. A. Sealove, “Peripheral Artery Disease: Current Insight Into the Disease and Its Diagnosis and Management,” *Mayo Clin. Proc.*, vol. 85, no. 7, pp. 678–692, Jul. 2010, doi: 10.4065/mcp.2010.0133.
- [3] M. Schillinger and E. Minar, “Percutaneous Treatment of Peripheral Artery Disease,” *Circulation*, vol. 126, no. 20, pp. 2433–2440, Nov. 2012, doi: 10.1161/CIRCULATIONAHA.111.036574.
- [4] S. Hajibandeh, S. Hajibandeh, S. A. Antoniou, F. Torella, and G. A. Antoniou, “Treatment strategies for in-stent restenosis in peripheral arterial disease: a systematic review,” *Interact. Cardiovasc. Thorac. Surg.*, vol. 28, no. 2, pp. 253–261, Feb. 2019, doi: 10.1093/icvts/ivy233.
- [5] R. Chaudhari, J. J. Vora, and D. M. Parikh, “A Review on Applications of Nitinol Shape Memory Alloy,” in *Recent Advances in Mechanical Infrastructure*, Singapore, 2021, pp. 123–132. doi: 10.1007/978-981-33-4176-0_10.
- [6] K. Maleckis *et al.*, “Nitinol Stents in the Femoropopliteal Artery: A Mechanical Perspective on Material, Design, and Performance,” *Ann. Biomed. Eng.*, vol. 46, no. 5, pp. 684–704, May 2018, doi: 10.1007/s10439-018-1990-1.
- [7] E. D. Grech, “Percutaneous coronary intervention. I: History and development,” *BMJ*, vol. 326, no. 7398, pp. 1080–1082, May 2003, doi: 10.1136/bmj.326.7398.1080.
- [8] M. S. Kim and L. S. Dean, “In-stent restenosis,” *Cardiovasc. Ther.*, vol. 29, no. 3, pp. 190–198, Jun. 2011, doi: 10.1111/j.1755-5922.2010.00155.x.
- [9] Y. Zhou, *Principles and Applications of Therapeutic Ultrasound in Healthcare*. Boca Raton: CRC Press, 2015. doi: 10.1201/b19638.
- [10] E. White *et al.*, “Focused Ultrasound Foundation: 2022 State of the Field Report”, [Online]. Available: <https://www.fusfoundation.org/the-foundation/foundation-reports/#SOTF>
- [11] A. I. Farrer *et al.*, “Characterization and evaluation of tissue-mimicking gelatin phantoms for use with MRgFUS,” *J. Ther. Ultrasound*, vol. 3, no. 1, p. 9, Jun. 2015, doi: 10.1186/s40349-015-0030-y.
- [12] C. A. Mattson and C. D. Sorensen, *Product Development: Principles and Tools for Creating Desirable and Transferable Designs*. Cham: Springer International Publishing, 2020. doi: 10.1007/978-3-030-14899-7.
- [13] D. J. Despain, “Design Validation of a Multi-Stage Gradually Deploying Stent,” M.Sc., Brigham Young University, United States -- Utah, 2021. Accessed: Mar. 06, 2023. [Online]. Available: <https://www.proquest.com/docview/2601399953/abstract/9E186F2F2B824659PQ/1>
- [14] V. Kallem and N. J. Cowan, “Image Guidance of Flexible Tip-Steerable Needles,” *IEEE Trans. Robot.*, vol. 25, no. 1, pp. 191–196, Feb. 2009, doi: 10.1109/TRO.2008.2010357.

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. DO NOT EXCEED FIVE PAGES.

NAME: Dillon, Christopher R

eRA COMMONS USER NAME (credential, e.g., agency login): crdillon

POSITION TITLE: Assistant Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	END DATE MM/YYYY	FIELD OF STUDY
Brigham Young University, Provo, UT	BS	04/2009	Mechanical Engineering
University of Utah, Salt Lake City, UT	PHD	08/2014	Bioengineering
University of Utah, Salt Lake City, UT	Postdoctoral Fellow	12/2017	MRI-guided Focused Ultrasound

A. Personal Statement

Over the past 15 years, I have developed expertise in focused ultrasound thermal therapies, tissue property characterization, tissue-mimicking phantoms, and ex vivo blood flow models that have positioned me to successfully lead the work proposed in this study. As a NIH F32 postdoctoral fellow, I developed procedures that enabled thorough characterization of uterine fibroid tissues, including the re-perfusion of those tissues and treatment with focused ultrasound. In my graduate research, I developed techniques using magnetic resonance-guided focused ultrasound (MRgFUS) data to noninvasively characterize tissue thermal, blood flow, and acoustic properties using phantoms and ex vivo porcine kidneys in addition to in vivo animal studies. The focused ultrasound, phantom fabrication, temperature control, and experimental methods proposed for these projects are in areas I have focused on in the past year as I have established my research laboratory and I am confident that our lab is well-positioned to fulfill the objectives of the proposed research.

1. Dillon C, Rezvani M, McLean H, Adelman M, Dassel M, Jarboe E, Janát-Amsbury M, Payne A. A tissue preparation to characterize uterine fibroid tissue properties for thermal therapies. Med Phys. 2019 Aug;46(8):3344-3355. PubMed Central PMCID: PMC6692230.
2. Dillon CR, Rieke V, Ghanouni P, Payne A. Thermal diffusivity and perfusion constants from in vivo MR-guided focussed ultrasound treatments: a feasibility study. Int J Hyperthermia. 2018 Jun;34(4):352-362. PubMed PMID: 28595499.
3. Dillon CR, Vyas U, Payne A, Christensen DA, Roemer RB. An analytical solution for improved HIFU SAR estimation. Phys Med Biol. 2012 Jul 21;57(14):4527-44. PubMed Central PMCID: PMC3402042.

B. Positions, Scientific Appointments and Honors**Positions and Scientific Appointments**

2021 -	Assistant Professor, Brigham Young University, Mechanical Engineering, Provo, UT
2018 - 2021	R&D S&E Computer Scientist, Sandia National Laboratories, Albuquerque, NM
2014 - 2017	Postdoctoral Research Associate, University of Utah, Utah Center for Advanced Imaging Research, Salt Lake City, UT
2009 - 2014	Graduate Research Assistant, University of Utah, Bioheat Transfer Laboratory, Salt Lake City, UT
2008 - 2009	Research Assistant, Brigham Young University, Friction Stir Welding Laboratory, Provo, UT

Honors

2021	Employee Recognition Award, Sandia National Laboratories
2017	Outstanding Trainee Presentation Award, 28th Annual UCAIR Symposium
2017	Higher Education Teaching Specialist, University of Utah

2015 - 2017	F32 Kirschstein-NRSA Postdoctoral Fellowship, National Institutes of Health
2014	Young Investigator Award, Focused Ultrasound Foundation
2014	New Investigator Travel Award, Society for Thermal Medicine
2008	Mechanical Engineering Department Scholarship, Brigham Young University
2001 - 2007	Robert C. Byrd Honors Scholarship, Utah State Office of Education
2001 - 2007	Gordon B. Hinckley Presidential Scholarship, Brigham Young University
2001 - 2007	National Merit Scholarship, Brigham Young University

C. Contribution to Science

1. I have developed two novel techniques for quantifying blood flow parameters for thermal modeling of MRgFUS treatments. Because blood flow draws heat from the target region and increases treatment unpredictability, accurate heat transfer modeling of blood flow for thermal therapies is a significant and critical challenge. The techniques I have developed are a key step to eliminating the uncertainty and unpredictability that blood flow brings to thermal therapies. These techniques may prove useful down the road in evaluating the appropriate patient population for our slowly deploying stents. The experiments performed for these past studies required phantoms, excised tissues, peristaltic pump-driven flows, and focused ultrasound heating that are foundational to the studies of the proposed work.
 - a. Shi YC, Parker DL, Dillon CR. Sensitivity of tissue properties derived from MRgFUS temperature data to input errors and data inclusion criteria: ex vivo study in porcine muscle. *Phys Med Biol*. 2016 Aug 7;61(15):N373-85. PubMed Central PMCID: PMC4970232.
 - b. Dillon CR, Borasi G, Payne A. Analytical estimation of ultrasound properties, thermal diffusivity, and perfusion using magnetic resonance-guided focused ultrasound temperature data. *Phys Med Biol*. 2016 Jan 21;61(2):923-36. PubMed Central PMCID: PMC4879616.
 - c. Dillon C, Roemer R, Payne A. Magnetic resonance temperature imaging-based quantification of blood flow-related energy losses. *NMR Biomed*. 2015 Jul;28(7):840-51. PubMed Central PMCID: PMC4510856.
2. As a postdoctoral fellow, I contributed to the validation of the Hybrid Angular Spectrum (HAS) method for predicting focused ultrasound power deposition in both simulation and experimental work. I also utilized HAS in studies evaluating how applying phase aberration correction techniques might improve accuracy and efficiency in MRgFUS therapies. Those projects utilized custom phantom fabrication similar to those that will be created in the proposed work. Finally, I assisted in developing a peristaltic pump-driven, temperature-controlled, conformable skin-cooling system for MRgFUS treatments that is now in clinical use at Stanford University. Each of these efforts has strengthened my understanding of the experimental and eventual clinical challenges associated with the proposed work and poised me to successfully address those challenges.
 - a. Merrill R, Odéen H, Dillon C, Bitton R, Ghanouni P, Payne A. Design and evaluation of an open-source, conformable skin-cooling system for body magnetic resonance guided focused ultrasound treatments. *Int J Hyperthermia*. 2021;38(1):679-690. PubMed Central PMCID: PMC8925859.
 - b. Johnson SL, Christensen DA, Dillon CR, Payne A. Validation of hybrid angular spectrum acoustic and thermal modelling in phantoms. *Int J Hyperthermia*. 2018;35(1):578-590. PubMed Central PMCID: PMC6365205.
 - c. Dillon CR, Farrer A, McLean H, Almquist S, Christensen D, Payne A. Experimental assessment of phase aberration correction for breast MRgFUS therapy. *Int J Hyperthermia*. 2018 Sep;34(6):731-743. PubMed PMID: 29278946.
 - d. Farrer AI, Almquist S, Dillon CR, Neumayer LA, Parker DL, Christensen DA, Payne A. Phase aberration simulation study of MRgFUS breast treatments. *Med Phys*. 2016 Mar;43(3):1374-84. PubMed Central PMCID: PMC4769272.

NAME: Jeffrey R Hill

POSITION TITLE & INSTITUTION: Associate Professor, Brigham Young University

A. PROFESSIONAL PREPARATION - (see [PAPPG Chapter II.C.2.f.\(i\)\(a\)](#))

INSTITUTION	LOCATION	MAJOR/AREA OF STUDY	DEGREE (if applicable)	YEAR (YYYY)
Pennsylvania State University	State College, PA	Mechanical Engineering	PhD	2011
Brigham Young University	Provo, UT	Mechanical Engineering	MS	2005
Brigham Young University	Provo, UT	Mechanical Engineering	BS	2004

B. APPOINTMENTS - (see [PAPPG Chapter II.C.2.f.\(i\)\(b\)](#))

From - To	Position Title, Organization and Location
2021 - Current	Associate Professor, Brigham Young University, Provo, UT
2016 – 2021	Principle Member of the Technical Staff, Sandia National Laboratories, Albuquerque, NM
2011 – 2016	Senior Member of the Technical Staff, Sandia National Laboratories, Albuquerque, NM
2008 – 2011	Visiting Research Investigator, University of Michigan, Ann Arbor, MI

C. PRODUCTS - (see [PAPPG Chapter II.C.2.f.\(i\)\(c\)](#)) Products Most Closely Related to the Proposed Project

J. Hill, K.W. Wang, J. Roh. "Position control of shape memory alloy actuators with load and frequency dependent hysteresis characteristics" Modeling, Signal Processing, and Control for Smart Structures, SPIE, vol. 7286, pp 74-82, 2009.

J. Hill, K. W. Wang, and H. Fang, "Advances of Surface Control Methodologies for Flexible Space Reflectors", Journal of Spacecraft and Rockets, vol. 50, no. 4, pp. 816–828, 2013.

J. Hill, K.W. Wang, "Development of the En Masse Elimination Algorithm for Actuator Grouping Optimization", Journal of Intelligent Material Systems and Structures, vol. 22, no. 11, pp 1203-1212, 2011.

J Kim, J. Hill, and K.W. Wang, "An Asymptotic Approach for the Analysis of Piezoelectric Fiber Composite Beams", Smart Materials and Structures, vol 20, no. 2, 2011.

J. Hill, J. Blotter, T. Leishman, and S. Sommerfeldt, "Development of a Weatherproof Windscreen for a Microphone Array", Noise Control Engineering Journal, Vol 54, n 3, p 169-176, May/June 2006. Sandia Internal Publications (restricted access)

Other Significant Products, Whether or Not Related to the Proposed Project

J. Hill, "21G030 Hypersonic Impact Fuzing". SAND2020-5576PE

J. Wise, G. Clark, S. Montgomery, J. Hill, "Dynamic PZT and Metal Oxide Varistor Material Characterization and Associated Device Response Data for Component Modeling, Design, and Validation Applications", SAND2020-8306

J. Hill, J. Dye, S. Curtis, R. McEntire, "Layer Detection for Embedded G-Switch", SAND2018-0699A

J. Hill, "G-Switch Past and Future Work", SAND2017-7978PE

J. Hill, "Layer Detection for Embedded G-Switch", SAND2017-13320

J. Hill, "W88 Impact Fuze Subassembly Reuse Studies", SAND2017-1342 J. Hill, "Using Multiple G-Switches for Target Detection", SAND2016-3763C

J. Hill, R. McEntire, "Using Multiple G-Switches for Target Detection", SAND2015-8360C

J. Hill, R. McEntire, "Tunability of G-Switches for Target Detection", SAND2015-5244C

J. Hill, R. McEntire, J. Kuhn, "Characterization of G-Switches for Target Detection", SAND2014-18796

D. SYNERGISTIC ACTIVITIES - (see [PAPPG Chapter II.C.2.f.\(i\)\(d\)](#))

ABET Program Evaluator, 2016 - Current